RSI Mean Reversion (BB) Initial Backtesting Report

Akylai Batyrbekova

1. Strategy overview 2
2. Strategy description 3
2.1 Indicators 3
2.2 Opening a position 3
2.3 Closing a position 4
3. Backtesting 4
3.1 Original strategy performance 4
3.2 Reproduced strategy performance & VGT Backtest 4
3.3 Hyperparameters 8
3.4 Optimization 8
4. Conclusion 9
5. Bibliography 9

Data source: Yahoo finance

Asset: VGT ETF (Vanguard Information Technology)

Timeframe: Daily; 2008-2018; 2019-2024; 20 ma, window=6

1. Strategy overview

The Mean Reversion strategy assumes that the price of a stock will eventually revert to their long-term average levels. Similar to the behaviour of a rubber band, stretch too far out and it will snap back.

A key characteristic of the Mean Reversion strategy is that it profits mostly in sideways markets and loses mostly in trending markets. Because the market is generally mean reverting, by simply going long when the price has come down too low or going short* when the price has come up too high, it's possible to find profitable trades more often.

This strategy utilizes the Relative Strength Index (RSI) and Simple Moving Average (SMA) to identify potential mean reversion opportunities in the market. When the RSI is below the buy threshold and the price is below the SMA, a buy signal is generated. When the RSI is above the sell threshold and the price is above the SMA, a sell signal is generated. The strategy also sets stop loss and profit target levels to manage trading risks and lock in profits.

The selection of an asset to trade using mean reversion is dependent on various factors such as market conditions, the entity's trading and investing expertise, and risk tolerance. Some commonly traded assets well-suited for mean reversion strategies include stocks, forex, commodities, exchange-traded funds (ETFs), and fixed income instruments.

Strategy Advantages

- Mean reversion strategies can capture reversal opportunities when prices deviate too far from their mean, potentially generating profits.
- Using the RSI indicator can effectively identify overbought and oversold conditions, improving the reliability of trading signals.
- Combining the SMA as a price benchmark can filter out some noise signals and enhance the quality of trades.
- Setting stop loss and profit target levels can effectively manage trading risks and protect account funds.

Strategy Risks

- Mean reversion strategies may not perform well in trending markets, as prices may continue to deviate from the mean without reverting.
- The choice of RSI and SMA parameters can impact strategy performance, and improper parameter settings may lead to false signals and losses.
- Fixed percentage stop loss and profit targets may not adapt well to different market volatility conditions, leading to premature stops or insufficient profit maximization.

2. Strategy description

2.1 Indicators

Since the mean reversion strategy relies on accurately betting that prices will revert back to its mean, we will need a combination of statistical factors to measure how far the current price has deviated and signal when the price has extremely high probability of reverting.

After searching around, I found 2 popular indicators:

- **Bollinger Bands:** The Bollinger bands technical analysis indicator consists of a middle band (simple moving average) and two outer bands calculated using the standard deviation. Prices are expected to revert to the middle band. Bollinger Bands plot a standard deviation above and below a simple moving average of the price. These 2 can be used as parameters and the combinations may be adjusted accordingly. I used the default values of a 20-day SMA and 2 standard deviations for this project.
- **Relative Strength Index (RSI):** The RSI ranges from 0 to 100 and is used to identify overbought and oversold conditions. An RSI above 70 suggests overbought conditions, while below 30 suggests oversold conditions, both implying a potential mean reversion. The average gain or loss used in the calculation is the average percentage gain or loss during a look-back period.

I chose a shorter 6-day period instead of the more commonly used 14-day period.

2.2 Opening a position

- Buy when the 6-period RSI falls below 30 & Price closes below the lower Bollinger band. Indicates that the price has fallen extremely low and that the stock is being oversold.

2.3 Closing a position

- Sell when the 6-period RSI rises above 70 & Price closes above the upper Bollinger band. Indicated that the price has come up extremely high and that the stock is being overbought.

3. Backtesting

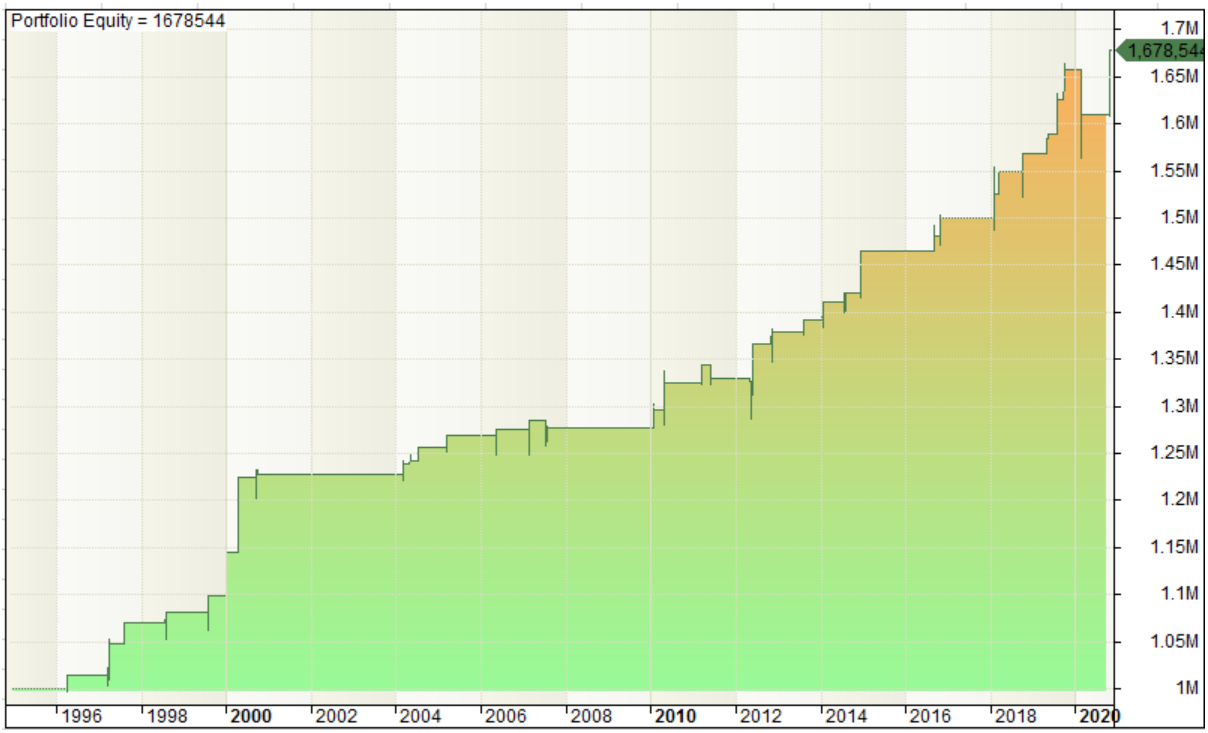
3.1 Original strategy performance

Number of trades: 38

Winning rate: 86.84%

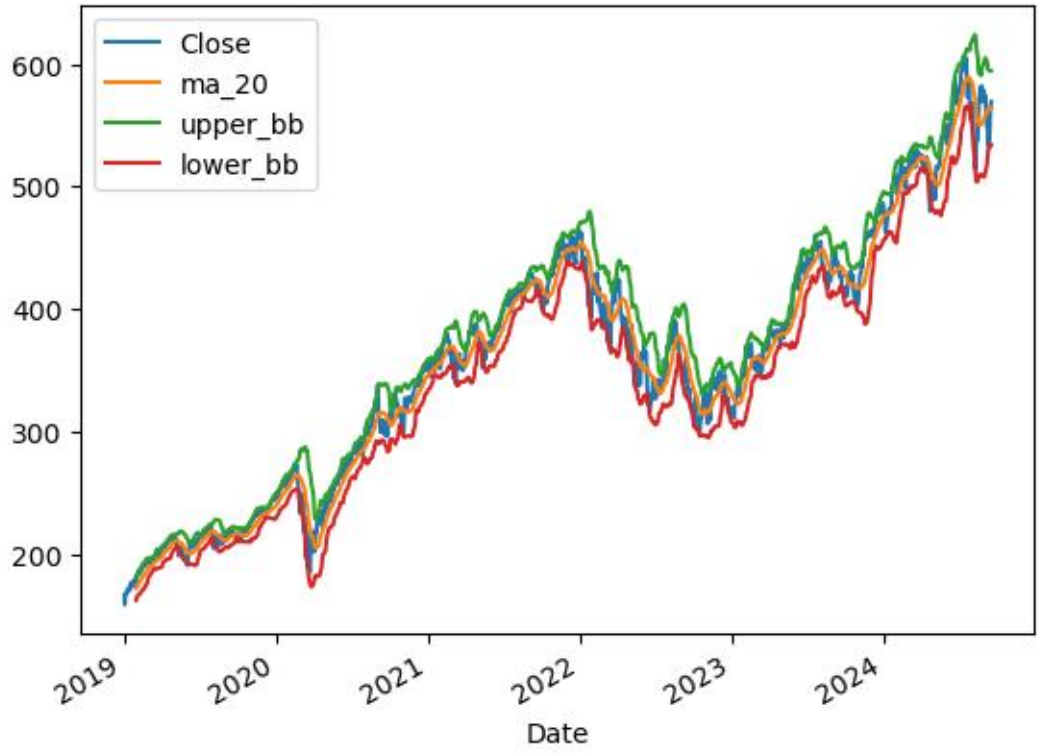
Average profit: 1.39%

Equity curve over the last 25 years:



3.2 Reproduced strategy performance & VGT Backtest

Period 01/01/2019 - 09/13/2024



Profits overview:

0	0.099248
1	0.069922

2	0.014087
3	0.131925
4	0.071583
5	0.043017
6	0.043024
7	-0.143756
8	-0.049103
9	0.078474
10	0.074227
11	0.034492
12	0.068232
13	0.042822

dtype: float64

Profit: 0.7107733639078471

Mean Reversion Statistics

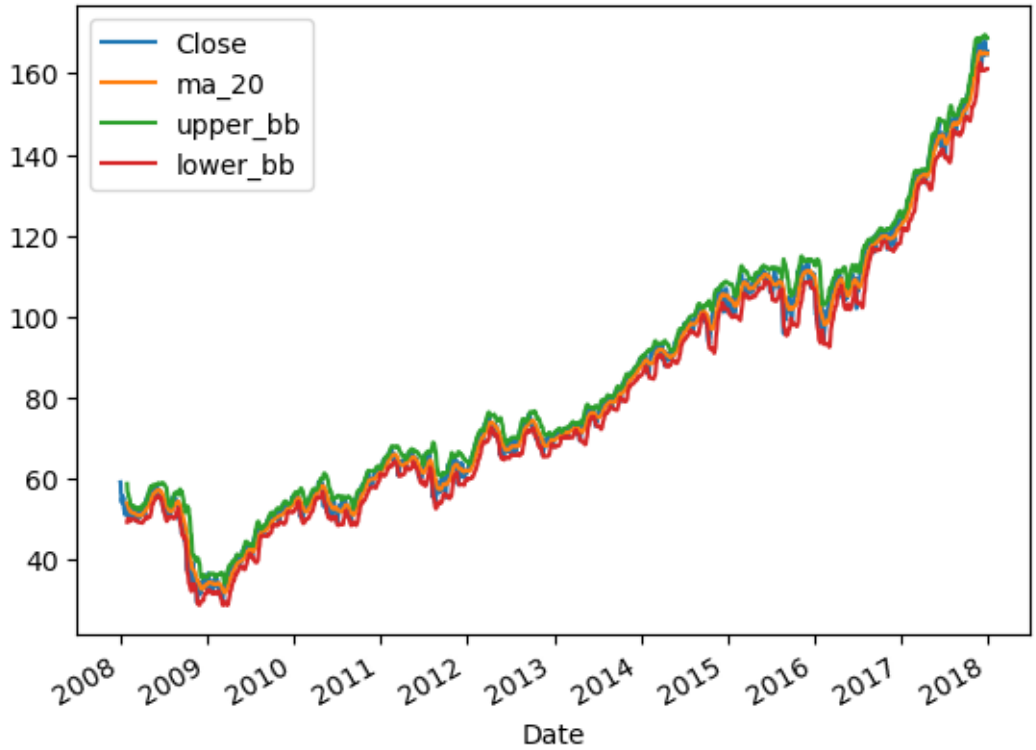
Number of trades:	14
Number of gains:	12
Number of losses:	2
Total return:	0.710773
Win rate:	0.857143
Average gain:	0.064254
Average loss:	-0.096429
Max return:	0.131925
Max loss:	-0.143756

The RSI mean reversion strategy applied to the VGT ETF over the period from 2019 to 2024 appears to be quite effective. With a high win rate of 85.71% and a total return of 71.08%, the strategy shows strong potential for profitability. However, it’s important to note that the average loss per trade is higher than the average gain, which suggests that while the strategy wins more often than it loses, the losses can be relatively larger when they do occur. It's also shown by a low Sharpe Ratio (0.23), suggesting that the returns were not very high relative to the risk taken. This could imply a need for careful risk management to mitigate the impact of occasional larger losses. Overall, the strategy seems promising but would benefit from further refinement to balance the risk-reward ratio.

Both strategies have a high winning rate, with the original slightly higher. The reproduced strategy shows a significant total return over the period. The reproduced strategy may require better risk management to handle larger losses and improve the risk-reward ratio. Both strategies show potential, but the original strategy has a slightly higher winning rate and a specified average profit per trade. The reproduced strategy, while effective in terms of total return, highlights the need for improved risk management due to larger losses and a low

Sharpe Ratio. Further refinement could help balance the risk-reward ratio for the reproduced strategy.

Period 01/01/2019 - 09/13/2024



Profit: 1.1671340226735207

3.3 Hyperparameters

- Adding stop loss condition

3.4 Optimization

- Sell when I hold a position and either receive a 'Sell' signal or the price drops below 95% of the last buy price.

Profit: 0.02445867233974397

Adding the stop loss condition did not improve profitability. The strategy already has a high winning rate, meaning most trades are successful. The stop loss might be closing trades prematurely, preventing potential profits. The stop loss could be too sensitive to short-term market volatility, triggering temporary price dips and resulting in unnecessary losses. The original strategy might have a better balance between risk and reward without the stop loss. While the stop loss aims to limit losses, it might also limit potential gains by closing trades too early. Given the high winning rate and the observed results, it might be more effective to focus on other risk management techniques or further refine the existing strategy without the stop loss.

4. Conclusion

In conclusion, the mean reversion strategy has the potential to deliver consistent returns in range-bound and volatile environments. However, it might not be the best strategy to implement in trending markets. It is also highly vulnerable to a stock market crash.

The high winning rate of this strategy suggests that most trades are successful, and the stop loss condition might be prematurely closing trades that would otherwise be profitable. Additionally, the stop loss could be too sensitive to short-term market fluctuations, leading to unnecessary losses. The original strategy appears to have a better risk-reward balance without the stop loss, indicating that the stop loss condition is not necessary and may even hinder performance.

5. Bibliography

1) https://www.tradingwithrayner.com/mean-reversion-trading-strategy/
2) https://www.linkedin.com/pulse/algorithmic-trading-mean-reversion-using-python-bryan-chen/
3) https://www.quantifiedstrategies.com/rsi-mean-reversion-trading-strategy/
4) https://fxopen.com/blog/en/mean-reversion-trading-strategies-and-indicators/
5) https://www.youtube.com/watch?v=AXc1YAsCdu